

SMART Framework

EVIDENCE ARCHITECTURE & STUDENT ONBOARDING SPECIFICATION

1. OBJECT STRUCTURE SCHEMA

Unified data models governing domain skill hierarchies, candidate skill claims, multi-source evidence verification, and dynamic readiness profiles.

Candidate

Candidate

- candidateId
- profile
- domain
- targetRole
- selectedSkills[]

Domain

Domain

- domainId
- name
- description
- skillIds[]
- roleIds[]

Role

Role

- roleId
- name
- domainId
- recommendedSkillIds[]
- optionalSkillIds[]

Skill

Skill

- skillCode
- name
- domain
- category
- prerequisites[]
- competencyModel
- proficiencyDefinitions
- assessmentBlueprint
- evidenceRequirements
- interviewBlueprint
- freshnessPolicy
- roleMappings[]

Evidence

Evidence

- evidenceId
- evidenceType
- source
- provenance
- evidenceDate
- freshness
- claim
- skillIds[]
- evidenceStrength
- evidenceReliability
- verificationStatus
- artifactIds[]
- contradictions[]

Work Experience

WorkExperience

- experienceId
- employer
- jobTitle
- employmentType
- startDate
- endDate
- department/team
- responsibilities[]
- deliverables[]
- skillMappings[]
- responsibilityLevels[]
- verifier
- verificationStatus

Competency

Competency

- └ competencyId
- └ skillCode
- └ capability
- └ observableBehaviours[]
- └ difficulty
- └ assessmentCriteria

Skill Claim

SkillClaim

- └ claimId
- └ candidateId
- └ skillCode
- └ targetProficiency
- └ status
- └ assessmentResult
- └ evidenceIds[]
- └ interviewId
- └ confidence
- └ finalProficiency

Project

Project

- └ projectId
- └ title
- └ projectType
- └ dates
- └ teamSize
- └ candidateRole
- └ problem
- └ approach
- └ stack/tools[]
- └ components[]
- └ personalContribution[]
- └ decisions[]
- └ constraints[]
- └ outcomes[]
- └ artifacts[]
- └ skillMappings[]
- └ verificationStatus

Certification / License

Credential

- └ credentialId
- └ issuer
- └ credentialName
- └ credentialType
- └ externalCredentialId
- └ issueDate
- └ expiryDate
- └ jurisdiction/scope
- └ verificationSource
- └ status
- └ assessmentType
- └ practicalComponent
- └ coveredSkills[]
- └ applicationEvidence[]

Passive Signal

PassiveSignal

- └ signalId
- └ source
- └ accountReference
- └ activity[]
- └ activityPeriod
- └ relevantArtifacts[]
- └ skillMappings[]
- └ signalStrength
- └ reliability

Artifact, Assessment & Verification

Artifact

└─ artifactId | type | reference | owner | createdAt
└─ relatedSkill | relatedContribution | integrityStatus

Assessment

└─ assessmentId | skillCode | blueprintVersion
└─ attemptId | result | competencyEvidence[] | timestamp

Interview

└─ interviewId | skillCode | questions[]
└─ evidenceTargets[] | assessmentTargets[] | responses[] | result

VerificationDecision

└─ decisionId | claimId | evidenceSummary | assessmentSummary
└─ interviewSummary | decision | confidence | reasons[] | reviewer

2. NEW & MODIFIED ONBOARDING FLOW

1. STUDENT & DOMAIN SELECTION

2. TARGET ROLE SELECTION

3. SMART RECOMMENDS SKILLS

4. STUDENT EDITS / CONFIRMS SKILLS

5. EVIDENCE INGESTION

- Work Experience
- Projects
- Certifications / Licenses
- Optional Passive Signals

6. SMART EXTRACTS CLAIMS

- Skills, Activities, Personal Contribution, Responsibility
- Decisions, Constraints, Outcomes, Evidence Gaps

7. EVIDENCE RECONCILIATION

- Cross-source consistency checking
- Missing evidence & contradiction detection

8. SHORT DIAGNOSTIC ASSESSMENT

9. CONFIDENCE-GAP ANALYSIS

- [Sufficient Confidence] → Fast-track to Verification
- [Insufficient / Contradictory] → Targeted Verification Pipeline:

10. VERIFICATION DECISION VERIFIED PROVISIONAL FAILED

11. VERIFIED SKILL CLAIM (Proficiency & Confidence Metrics)

12. ROLE READINESS INDEX

13. CONTINUOUS PROFILE

Core Onboarding Principles

Anti-Fatigue Mechanics

- Evidence is reused across multiple skills automatically.
- Students never repeat the entire process for every skill.
- Defensive interviews are triggered *only when necessary*.
- Assessments are adaptive/diagnostic rather than unnecessarily long.

Anti-Gaming Safeguards

- Student provides raw facts and evidence; SMART determines proficiency.
- No single certificate, project, GitHub profile, or self-report independently establishes Advanced/Professional level.
- Important claims are cross-checked across independent evidence sources.

3. EVIDENCE COLLECTION FIELDS

A. Universal Evidence

Evidence

```
├─ evidenceId
├─ evidenceType
├─ source
├─ sourceOwner
├─ sourceReference
├─ evidenceDate
├─ submissionDate
├─ relatedSkillIds[]
├─ claim
├─ context
├─ provenance
├─ accessibility
└─ verificationStatus
```

B. Candidate Contribution

Prevents students from claiming an entire project/job as their own.

Contribution

```
├─ whatWasDone
├─ personalContribution
├─ responsibilityLevel
│   └─ Observed
│   └─ Assisted
│   └─ Implemented
│   └─ Owned
│   └─ Designed / Decided
├─ independenceLevel
├─ frequency
├─ scope
└─ decisionsMade[]
```

C. Activity / Work

Activity

- └─ task
- └─ objective
- └─ tools / methods
- └─ process
- └─ constraints
- └─ decision
- └─ reasonForDecision
- └─ result
- └─ measurableOutcome
- └─ baseline
- └─ finalMetric

Core Sequence: *What did you do? → Why did you do it? → What did you decide? → Why? → What happened?*

D. Artifacts

Artifact

- └─ artifactType
- └─ artifactReference
- └─ ownership
- └─ createdAt
- └─ relatedComponent
- └─ relatedSkill
- └─ contributionSupported
- └─ accessibility
- └─ integrityStatus

Supported Formats: Code repositories, designs, reports, research papers, CAD files, financial models, presentations, datasets, deployments, portfolios.

4. WORK EXPERIENCE & 5. PROJECT SCHEMA

4. Work Experience Fields

WorkExperience

- └─ employer | jobTitle | employmentType
- └─ startDate | endDate | location
- └─ department/team
- └─ responsibilities[] | deliverables[]
- └─ skillMappings[] | personalContributions[]
- └─ responsibilityLevels[]
- └─ verifierName | verifierRole | verifierContact
- └─ employmentVerification

Responsibility (Per Item)

- └─ task | skill | personalContribution
- └─ responsibilityLevel | independence
- └─ tools | decision | constraint
- └─ outcome | artifact

5. Project Fields

Project

- └─ title | projectType | dates | teamSize
- └─ candidateRole | problem | targetUsers
- └─ requirements | approach | stack/tools
- └─ components | personalContribution
- └─ decisions | constraints | outcomes
- └─ teamContribution | artifacts | NDASatus

ProjectSkillMapping (Per Skill)

- └─ skill | specificContribution
- └─ componentWorkedOn | actionsPerformed
- └─ decisionsMade | constraintsHandled
- └─ artifact | verificationStatus

SMART Disambiguation Logic

Prevents unwarranted attribution: Instead of assuming "The project used AWS, therefore candidate knows AWS", SMART demands: "What did you personally do with AWS?"

6. CREDENTIALS, 7. PASSIVE SIGNALS & 8. VERIFICATION METADATA

6. Certification / License

Credential

- | issuer | credentialName | credentialType
- | externalCredentialId | issueDate | expiryDate
- | jurisdiction | scope | verificationSource
- | credentialStatus | assessmentType
- | practicalComponent | coveredTopics[] | coveredSkills[]

ApplicationEvidence

- | claimedApplication | linkedProject
- | linkedWorkExperience | artifact
- | corroborationStatus

Separation Rule: *Credential authenticity ≠ practical competency.*

7. Passive Signal Fields

PassiveSignal

- | source (GitHub, LeetCode, HackerRank, e
- | accountReference | activityType
- | activityPeriod | relevantArtifacts
- | skillMappings | signalStrength
- | reliability | anomalies

8. Verification Metadata

VerificationMetadata

- | verificationMethod | verifiedBy
- | verificationDate | evidenceStrength
- | evidenceReliability | contradictions[]
- | linkedEvidenceIds[] | freshnessClass
- | reviewRequired | auditTrail

The Fundamental Verification Axiom

Don't ask: "How much evidence does this student have?"

Ask: "How directly does this evidence support this specific claim, how reliable is its source, and is it consistent with everything else we know?"

This architectural framework universally scales across software engineering, finance, mechanical engineering, healthcare, management, design, marketing, research, and technical domains without altering the underlying verification engine.